Methods 8

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Apologies for the late submission! You can view my repo's commit history here if that helps to verify that this is my own work done independent from the solutions.

1.

a.

$$\text{Provided that :} \qquad \bar{Y}(A=0,B=0)=1, \qquad \bar{Y}(A=1,B=0)=5,$$

$$\bar{Y}(A=0,B=1)=4, \qquad \bar{Y}(A=1,B=1)=7$$

we can hold the following to be true.

$$\begin{aligned}\bar{Y}(A=0,B=0) &= \mathbb{E}(Y_i \mid A=0,B=0)\\ &= \mathbb{E}(\beta_0+\beta_1\cdot 0+\beta_2\cdot 0+\beta_3\cdot 0+\epsilon_i)\\ &= \beta_0+0+0+0+\mathbb{E}(\epsilon_i)\\ &= \beta_0+0=1\\ &\therefore \beta_0=1\end{aligned}$$

$$\begin{aligned}\bar{Y}(A=1,B=0) &= \mathbb{E}(Y_i \mid A=1,B=0)\\ &= \mathbb{E}(\beta_0+\beta_1\cdot 1+\beta_2\cdot 0+\beta_3\cdot 0+\epsilon_i)\\ &= \beta_0+\beta_1+0+0+\mathbb{E}(\epsilon_i)\\ &= 1+\beta_1+0=5\\ &\therefore \beta_1=4\end{aligned}$$

$$\begin{aligned}
\bar{Y}(A=0,B=1) &= \mathbb{E}(Y_i \mid A=0,B=1) \\
&= \mathbb{E}(\beta_0 + \beta_1 \cdot 0 + \beta_2 \cdot 1 + \beta_3 \cdot 0 + \epsilon_i) \\
&= \beta_0 + 0 + \beta_2 + 0 + \mathbb{E}(\epsilon_i) \\
&= 1 + \beta_2 + 0 = 4 \\
&\therefore \beta_2 = 3
\end{aligned}$$

$$\begin{aligned}
\bar{Y}(A=1,B=1) &= \mathbb{E}(Y_i \mid A=1,B=1) \\
&= \mathbb{E}(\beta_0 + \beta_1 \cdot 1 + \beta_2 \cdot 1 + \beta_3 \cdot 1 + \epsilon_i) \\
&= \beta_0 + \beta_1 + \beta_2 + \beta_3 + \mathbb{E}(\epsilon_i) \\
&= 1 + 4 + 3 + \beta_3 = 7 \\
&\therefore \beta_3 = -1
\end{aligned}$$

Final Answer: $\boxed{\beta_0 = 1, \beta_1 = 4, \beta_2 = 3, \beta_3 = -1}$

b.

In order to get estimates for the four regression coefficients, we need estimates for the four averages above. We have two variables, A and B, which can take values 0 or 1 depending on whether or not the value observed was at a "high" or "low" level for each. β_0, the intercept or "baseline", necessitates that we have a treatment group where $A = 0, B = 0$. Then, by comparing to our baseline, we can get our treatment effects for A and B by changing the level one at a time: one group is $A = 1, B = 0$, and the other $A = 0, B = 1$. This lets us estimate β_1 and β_2. We also have to estimate the interaction effect β_3, so we need treatment group $A = 1, B = 1$. This constructs our full factorial design, where we have 2^2 treatment groups, one for each combination of treatment levels. Then, our observational units are randomly assigned to evenly split among the treatment groups.

In summary:

Group	Treatment	Number of units
Group 1	$A = 0, B = 0$	$n = 50$
Group 2	$A = 1, B = 0$	$n = 50$
Group 3	$A = 0, B = 1$	$n = 50$
Group 4	$A = 1, B = 1$	$n = 50$

c.

A full factorial design would be better:

We need to compare $A = 0,\ B = 0,\ C = 0$ and $A = 0,\ B = 0,\ C = 1$ to see C's treatment effect; i.e. how it changes the predicted result when all other variables are held constant. Then, to see how the effect changes for A and B, we need to compare $A = 0,\ B = 0,\ C = 1$ versus $A = 1,\ B = 0,\ C = 1$; likewise for B, we need to compare \$ A = 0, ~B = 0, ~C = 1\$ to $A = 0,\ B = 1,\ C = 1$. Assuming that our treatment groups from **(b)** had level $C = 0$, then this introduces 3 new groups. That gives us 7 treatment groups to estimate the interaction effects described.

A full factorial design would also include a three-way interaction effect, getting us $7 + 1 = 2^3$ groups, each group getting 25 units through random assignment. While a three-way interaction effect was not mentioned in the prompt, we also don't have reasonable certainty to assume that interaction effect to be zero. So, if we did a fractional factorial design, we would be makiing the dubious assumption that there is no three-way interaction effect, and we would also be missing the interactions between C and A or B, or we would be missing treatment effects.

d.

To consider treatment and interaction effects with C, we assume:

$$Y_i = \beta_0 + \beta_1 A_i + \beta_2 B_i + \beta_3 AB_i + \beta_4 C_i + \beta_5 AC_i + \beta_6 BC_i + \beta_7 ABC_i + \epsilon_i$$

With the assistant's proposed fractional factorial design, we can estimate $\bar{Y}(A = 1, B = 0, C = 0)$, $\bar{Y}(A = 0, B = 1, C = 0)$, $\bar{Y}(A = 0, B = 0, C = 1)$, and $\bar{Y}(A = 1, B = 1, C = 1)$ to estimate β_1, β_2, β_4, and β_7.

However, recall that the population averages gave us β_0, β_1 (treatment effect of a), β_2 (treatment effect of b), and β_3 (interaction effect between A and B). Therefore, in this fractional design, the interaction effect between A and B is aliased with the treatment effect of C. Therefore, we can't estimate the treatment effect of C.

e.

The design from part **(d)** assumes that the regression coefficients that it can't estimate (the experiment results can't estimate $\beta_0, \beta_3, \beta_5, \beta_6$) to be 0. This includes the interaction effect between A and B, β_3. So, if we assume $\beta_3 = 0$:

$$\begin{aligned}\bar{Y}(A=1,B=1) &= \beta_0 + \beta_1 + \beta_2 + \beta_3 \\ &= 1 + 4 + 3 + 0 \\ &= 8\end{aligned}$$

Therefore, to estimate treatment effect of C, we would require $\bar{Y}(A=1,B=1) = 8$.

2.

Run `FourFactorAnalysis.R`:

```
source("FourFactorAnalysis.R")
```

a.

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.1.4     v readr     2.1.5
v forcats   1.0.0     v stringr   1.5.1
v ggplot2   3.5.1     v tibble    3.2.1
v lubridate 1.9.3     v tidyr     1.3.1
v purrr     1.0.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
library(kableExtra)
```

```
Attaching package: 'kableExtra'

The following object is masked from 'package:dplyr':

    group_rows
```

```r
pairwise_lm = lm(y ~ a + b + c + d + a*b + a*c + a*d + b*c + b*d + c*d, data = thedata)

coeff_pop = c("Intercept" = trt0, "A" = trta, "B" = trtb, "C" = trtc, "D" = trtd, "AB" = trta
tibble(
  coefficient = names(coeff_pop)[-(12:16)],
  estimate = unname(pairwise_lm$coefficients),
  pop_value = unname(coeff_pop)[-(12:16)]
) |> kable(digits = 3)
```

coefficient	estimate	pop_value
Intercept	9.850	10
A	2.167	2
B	-0.861	-1
C	4.075	4
D	3.020	3
AB	3.884	4
AC	-1.143	-1
AD	2.040	2
BC	5.990	6
BD	-3.106	-3
CD	2.058	2

Looking at the outputs for the regression and for the true coefficients side by side, we can see that they are very similar (tend to only be off by $\approx \pm 0.3$ at most).